M13.4 Communication Batch 01 · Full Bank

Master-manual pages 1–117 · 324 questions · 177 from highlighted material (●).
Each topic: highlighted questions first, then the rest, each from Level 1 to Level 3. Answer key at the back, grouped by topic.

#	Topic	Pages	● High.	Total
1	Radio Technology Basics	6-7	7	11
2	AC Properties, Frequency and Angular Frequency	8-14	14	19
3	Harmonic, Triangular and Sawtooth Waves	15-20	5	10
4	Frequency Bands, Wavelength and Aviation Frequencies	21-31	24	36
5	Radio Wave Propagation: Ground Wave and Sky Wave	32-39	8	28
6	Ionosphere and Ionospheric Effects	40-47	20	41
7	Antenna Fundamentals, Fields, Polarization and Radiation Pattern	48-65	22	35
8	Antenna Types	66-82	26	42
9	Antenna Characteristics and Antenna Tuning	83-89	13	27
10	RF Lines, Matching and Standing Waves	90-103	26	40
11	Resonant Circuits	105-117	12	35

1 · Radio Technology Basics

Pages 6-7 · 11 questions

HIGHLIGHTED

1 **What are transmitting and receiving equipment basically made of?**

A Microphones, mixers and RF lines
B Electronic power supplies, amplifiers and oscillators
C Antennas, modulators and loudspeakers

Level 1 · Page 6

2 **What is the simplest radio communication system made of?**

A A transmitter, a receiver and a relay
B A transmitter and a receiver
C An oscillator and an antenna

Level 1 · Page 6

3 **Which transmitter stage places the audio intelligence on the RF signal?**

A The modulator
B The oscillator
C The RF amplifier

Level 1 · Page 6

4 **What converts sonic waves into electrical signals?**

A Microphones
B Oscillators
C Loudspeakers

Level 1 · Page 6

5 **What do loudspeakers convert?**

A Electrical signals into RF signals
B Sonic waves into electrical signals
C Electrical signals into sonic waves

Level 1 · Page 6

6 **How are the transmitter and receiver connected in a basic radio system?**

A By the medium the waves travel through
B By a matched RF line in all cases
C By the ground wave in all cases

Level 2 · Page 6

7 **Do electromagnetic waves need a medium to travel?**

A Yes, they need air to travel through
B No, no medium is necessary
C Yes, they need a wire or a cable

Level 2 · Page 6

NOT HIGHLIGHTED

8 **What does a transmitter consist of?**

A An amplifier, a detector and a loudspeaker
B A microphone, a loudspeaker and an antenna
C An oscillator, RF amplifiers and a modulator

Level 1 · Page 6

9 **What does the oscillator in a transmitter generate?**

A A basic radio frequency
B The modulated audio signal
C The audio intelligence

Level 1 · Page 6

10 **What does the receiving antenna turn the received energy into?**

A Sonic waves for the loudspeaker
B A flow of alternating RF currents
C A steady direct current

Level 2 · Page 6

11 **What does the receiver turn the RF currents into?**

A A new basic radio frequency
B Electromagnetic waves for the antenna
C The intelligence in the transmission

Level 2 · Page 6

Coverage All text on pages 6–7 covered. Page 7 is a figure only.

2 · AC Properties, Frequency and Angular Frequency

Pages 8-14 · 19 questions

HIGHLIGHTED

12 **Which types of current are used in communications engineering?**

A Both AC and DC
B AC only
C DC only

Level 1 · Page 8

13 **What is the time taken for one complete cycle called?**

A The frequency f
B The angular frequency ω
C The cycle time T

Level 1 · Page 8

14 **What is the symbol for the period?**

A T
B ω
C f

Level 1 · Page 8

15 **What unit is the period measured in?**

A Hertz (Hz)
B Seconds (s)
C Radians per second

Level 1 · Page 8

16 **How is the frequency of AC defined?**

A The number of half-waves in one second
B The time taken for one complete cycle
C The number of complete cycles in one second

Level 1 · Page 10

17 **What does 1 cps equal?**

A 1 Hz
B 1 kHz
C 1 s

Level 1 · Page 10

18 **What is the formula for angular frequency?**

A $\omega = 2\pi / f$
B $\omega = 2\pi \times f$
C $\omega = f / 2\pi$

Level 1 · Page 12

19 **How does the current behave in AC?**

A The current varies in quantity but not direction
B The current varies in direction but not quantity
C The current varies in direction and in quantity

Level 2 · Page 8

20 **How often does AC repeat its change of polarity and current?**

A At random, with the load
B Periodically, cycle after cycle
C Only once, at switch-on

Level 2 · Page 8

21 **What does one complete AC cycle consist of?**

A One positive alteration only
B Two positive alterations
C One positive and one negative alteration

Level 2 · Page 9

22 **How is frequency related to the cycle time T?**

A It is twice T
B It is the reciprocal of T
C It is equal to T

Level 2 · Page 10

23 **What does the rotational speed of a generator's conductor tip depend on?**

A The RPM and the diameter of the generator
B The diameter of the generator only
C The voltage and the current

Level 2 · Page 12

24 **What do the length and rotation of a phasor show?**

A Length shows the cycle time, rotation the phase
B Length shows the amplitude, rotation the frequency
C Length shows the frequency, rotation the amplitude

Level 2 · Page 12

25 **How does angular velocity differ from angular frequency?**

A Both of them are vector quantities
B Angular velocity is a scalar, angular frequency a vector
C Angular velocity is a vector, angular frequency a scalar

Level 2 · Page 13

NOT HIGHLIGHTED

26 **Where is AC used in power engineering?**

A In the mains supply
B In antenna feeders only
C In battery circuits only

Level 1 · Page 8

27 **What waveform does a generator with rotating parts produce?**

A A rectangular AC
B A sinusoidal AC
C A sawtooth AC

Level 1 · Page 8

28 **What unit is frequency measured in?**

A Seconds (s)
B Radians (rad)
C Hertz (Hz)

Level 1 · Page 10

29 **How does the voltage behave during the positive half-wave?**

A It rises from −peak to +peak
B It stays at the positive peak value
C It rises from 0 to +peak and falls back to 0

Level 2 · Page 8

30 **What happens to the cycle time when the frequency increases?**

A It becomes longer
B It becomes shorter
C It stays the same

Level 2 · Page 10

Coverage All text on pages 8–14 covered. Pages 11 and 14 are figures only. Page 9 figure labels used.
Conflict Page 9 (figure) uses the word 'Alteration' for each half of the cycle. Questions keep the LTT word.
Conflict Pages 12 and 13 both carry the LTT footer 'Page 8'; page 13 is an added page.

3 · Harmonic, Triangular and Sawtooth Waves

Pages 15-20 · 10 questions

HIGHLIGHTED

31 **What is the sine wave with the same frequency as a non-sinusoidal wave called?**

A A further harmonic or 2nd harmonic
B An odd harmonic or 3rd harmonic
C The basic oscillation or 1st harmonic

Level 2 · Page 16

32 **What are multiples of the basic frequency called?**

A Further harmonic waves
B Sawtooth waves
C Basic oscillations

Level 2 · Page 16

33 **What waveform do you get by superimposing the 1st, 3rd and 5th harmonics?**

A An almost triangular waveform
B An almost sawtooth waveform
C An almost rectangular waveform

Level 3 · Page 16

34 **Which harmonics are used to build a triangular wave?**

A The odd harmonics
B The even harmonics only
C Both odd and even harmonics

Level 3 · Page 19

35 **Which harmonics does a sawtooth wave need?**

A The odd harmonics only
B All harmonics, odd and even
C The even harmonics only

Level 3 · Page 19

NOT HIGHLIGHTED

36 **Which harmonics make up a rectangular wave?**

A A finite number of even harmonics
B The 1st and 2nd harmonics only
C An infinite number of uneven harmonics

Level 2 · Page 16

37 **What decides the harmonic numbers and amplitudes of a wave?**

A The frequency of the 1st harmonic
B The shape of the non-sinusoidal oscillation
C The amplitude of the supply voltage

Level 2 · Page 16

38 **How is a harmonic's amplitude found for a rectangular wave?**

A 1st harmonic amplitude ÷ harmonic number
B 1st harmonic amplitude ÷ harmonic number squared
C 1st harmonic amplitude × harmonic number

Level 3 · Page 16

39 **How fast do the harmonics of a triangular wave get smaller?**

A In direct proportion to the harmonic number
B With the cube of the harmonic number
C With the square of the harmonic number

Level 3 · Page 19

40 **How fast do the harmonics of a sawtooth wave get smaller?**

A In direct proportion to the harmonic number
B With the square of the harmonic number
C It does not decrease at all

Level 3 · Page 19

Coverage All text covered. Page 15 repeats page 16 (same text), cited as page 16. Pages 17, 18 and 20 are figures; page 18 caption used.

4 · Frequency Bands, Wavelength and Aviation Frequencies

Pages 21-31 · 36 questions

HIGHLIGHTED

41 **Which are the three most important frequency bands in communication engineering?**

A VLF, LF and MF
B HF, VHF and UHF
C LF, MF and HF

Level 1 · Page 21

42 **What is the frequency range of HF?**

A 30 MHz to 300 MHz
B 300 kHz to 3 MHz
C 3 MHz to 30 MHz

Level 1 · Page 21

43 **What is the frequency range of VHF?**

A 3 MHz to 30 MHz
B 30 MHz to 300 MHz
C 300 MHz to 3000 MHz

Level 1 · Page 21

44 **What is the frequency range of UHF?**

A 300 MHz to 3000 MHz
B 30 MHz to 300 MHz
C 3 GHz to 30 GHz

Level 1 · Page 21

45 **Which of these are communication uses of radio frequencies?**

A Electro-medicine, drying and heating
B Data communication, radio and satellite broadcast
C Inductive heating, annealing and melting

Level 1 · Page 22

46 **How fast does a wave travel in free space (vacuum)?**

A The speed of light c
B Half the speed of light
C The speed of sound

Level 1 · Page 23

47 **How is the wavelength λ calculated?**

A $\lambda = c \times T = c / f$
B $\lambda = c \times f$
C $\lambda = f / c$

Level 1 · Page 23

48 **What is the approximate speed of light in a vacuum?**

A 30 000 km per second
B 3 000 000 km per second
C 300 000 km per second

Level 1 · Page 23

49 **How are radio frequencies up to 300 GHz divided?**

A Into octave ranges
B Into decadic ranges
C Into equal 1 GHz ranges

Level 1 · Page 27

50 **What is each decade of radio frequencies called?**

A A channel
B A frequency band
C A radar band

Level 1 · Page 27

51 **How are the frequencies used for radar technology divided?**

A Into audio frequency bands
B Into decadic bands only
C Into special radar bands

Level 1 · Page 27

52 **What is the frequency range of VHF COM?**

A 108 MHz to 118 MHz
B 225 MHz to 400 MHz
C 118 MHz to 137 MHz

Level 1 · Page 28

53 **What is the frequency range of HF COM?**

A 2 MHz to 30 MHz
B 118 MHz to 137 MHz
C 3 kHz to 30 kHz

Level 1 · Page 28

54 **What is the frequency range of UHF COM?**

A 962 MHz to 1213 MHz
B 225 MHz to 400 MHz
C 118 MHz to 137 MHz

Level 1 · Page 28

55 **What is the frequency range of the ADF?**

A 200 kHz to 1750 kHz
B 10 kHz to 24 kHz
C 2 MHz to 30 MHz

Level 1 · Page 28

56 **What frequency does SatCom use?**

A 4.3 GHz
B 9.375 GHz
C 1.5 GHz

Level 1 · Page 28

57 **What frequency does weather radar use?**

A 8.8 GHz
B 9.375 GHz
C 4.3 GHz

Level 1 · Page 28

58 **What frequency range does VLF navigation use?**

A 10 kHz to 24 kHz
B 300 Hz to 3 kHz
C 200 kHz to 1750 kHz

Level 1 · Page 28

59 **What is the identification tone of the DME?**

A 1020 Hz
B 400 Hz
C 1350 Hz

Level 1 · Page 28

60 **What is the identification tone of VOR, ILS and ADF?**

A 1020 Hz
B 1350 Hz
C 3000 Hz

Level 1 · Page 28

61 **What frequency range does voice cover?**

A 300 Hz to 3 kHz
B 3 kHz to 30 kHz
C 20 Hz to 20 kHz

Level 1 · Page 28

62 **Which tones does the marker use?**

A 300, 1020 and 1350 Hz
B 1020, 1350 and 3000 Hz
C 400, 1300 and 3000 Hz

Level 1 · Page 28

63 **What does the propagation speed of a wave depend on?**

A The medium used
B The length of the antenna
C The amplitude of the voltage

Level 2 · Page 23

64 **What happens to the propagation speed as the density of the medium increases?**

A It stays the same
B It decreases
C It increases

Level 2 · Page 23

NOT HIGHLIGHTED

65 **What does IF stand for?**

A Isotropic Frequency
B Intermediate Frequency
C Internal Frequency

Level 1 · Page 27

66 **Which radar band covers 5200 MHz to 10 900 MHz?**

A S band
B C band
C X band

Level 1 · Page 27

67 **What is the frequency range of the L band?**

A 1550 MHz to 5200 MHz
B 390 MHz to 1550 MHz
C 3900 MHz to 6200 MHz

Level 1 · Page 27

68 **What is the frequency range of the localizer?**

A 108 MHz to 112 MHz
B 329.3 MHz to 399.95 MHz
C 112 MHz to 118 MHz

Level 1 · Page 28

69 **What is the frequency range of the VOR?**

A 108 MHz to 112 MHz
B 118 MHz to 137 MHz
C 112 MHz to 118 MHz

Level 1 · Page 28

70 **What is the frequency range of the glide slope?**

A 962 MHz to 1213 MHz
B 329.3 MHz to 399.95 MHz
C 108 MHz to 112 MHz

Level 1 · Page 28

71 **What frequency does the marker beacon use?**

A 75 MHz
B 440 MHz
C 108 MHz

Level 1 · Page 28

72 **What is the frequency range of the DME?**

A 962 MHz to 1213 MHz
B 329.3 MHz to 399.95 MHz
C 1030 MHz to 1090 MHz

Level 1 · Page 28

73 **What is the frequency range of ATC?**

A 4.25 GHz to 4.35 GHz
B 1030 MHz to 1090 MHz
C 962 MHz to 1213 MHz

Level 1 · Page 28

74 **What is the frequency range of the LRRA?**

A 1.2 GHz to 1.5 GHz
B 8.8 GHz to 9.375 GHz
C 4.25 GHz to 4.35 GHz

Level 1 · Page 28

75 **What rule of thumb gives the top of each frequency band?**

A Add 30 MHz to the lower limit
B Multiply the lower limit by 2
C Multiply the lower limit by 10

Level 2 · Page 21

76 **Which band uses both ground wave and sky wave?**

A VHF
B MF
C HF

Level 2 · Page 30

Coverage All text covered. Page 24 repeats page 23. Page 26 repeats the page 22 chart. Page 29 repeats the page 28 aviation frequencies as a picture. Page 30 table used for propagation types.
Conflict Page 27 radar band table: the C band (3900–6200 MHz) overlaps the S band (1550–5200 MHz) and the X band (5200–10 900 MHz). Questions use the table values as printed.
Conflict Page 28 (figure): 'ALTM 440 MHz' is highlighted but not explained anywhere; no question was written on it.
Conflict Page 31 is a general spectrum picture with no LTT text; not used for questions.

5 · Radio Wave Propagation: Ground Wave and Sky Wave

Pages 32-39 · 28 questions

HIGHLIGHTED

77 **How fast do radio waves travel?**

A Approximately the speed of sound
B Exactly half the speed of light
C Approximately the speed of light

Level 1 · Page 32

78 **Which part of the radio wave travels outward along the ground?**

A The space wave
B The ground wave
C The sky wave

Level 1 · Page 32

79 **Which part of the radio wave travels upward and outward?**

A The sky wave
B The ground wave
C The surface wave

Level 1 · Page 32

80 **What does the term wave propagation mean?**

A The direction of the electrical field
B The rate at which a wave loses energy
C The way waves are transmitted from one end to another

Level 1 · Page 34

81 **Which wave carries most daytime broadcast reception?**

A The sky wave
B The ground wave
C The space wave

Level 1 · Page 36

82 **What is the ground wave used for?**

A Long-range HF daylight communication only
B Short range at LF with low power, long range at HF with very high power
C Short range at HF with low power, long range at LF with very high power

Level 2 · Page 32

83 **What is the sky wave used for by day?**

A Long-range high frequency communication
B Line-of-sight UHF communication
C Short-range low frequency communication

Level 2 · Page 32

84 **Which frequencies only travel in a direct line?**

A From HF and above
B LF and VLF only
C MF and below

Level 2 · Page 34

NOT HIGHLIGHTED

85 **Which frequencies travel as space waves in a line of sight?**

A MF and lower
B HF only
C VHF and higher

Level 1 · Page 34

86 **Which frequencies follow the curvature of the earth as ground waves?**

A UHF only
B VHF and higher
C MF and lower

Level 1 · Page 34

87 **How much better does salt water conduct than dry soil?**

A 5000 times
B 50 000 times
C 500 times

Level 1 · Page 36

88 **What is the usual top frequency for ground wave propagation?**

A About 2 MHz
B About 300 MHz
C About 30 MHz

Level 1 · Page 37

89 **What frequency range is used for sky wave propagation?**

A 30 MHz to 300 MHz
B 3 MHz to 30 MHz
C 300 kHz to 3 MHz

Level 1 · Page 39

90 **What shape is the field pattern around a vertical antenna?**

A A huge doughnut lying on the ground
B A figure eight standing upright
C A narrow pencil beam

Level 2 · Page 32

91 **What does the sky wave provide at night?**

A Short-range contacts at higher frequencies
B Long-range contacts at somewhat lower frequencies
C Line-of-sight contacts at UHF

Level 2 · Page 32

92 **How does the ground wave lose energy as it travels?**

A It is refracted by the ionosphere
B It is reflected by the sea surface
C It induces eddy currents in the earth

Level 2 · Page 36

93 **What does increasing the frequency do to ground wave attenuation?**

A It has no effect on it
B It rapidly decreases it
C It rapidly increases it

Level 2 · Page 36

94 **Which frequencies do shore-based transmitters use for long-range ground waves?**

A 18 kHz to 300 kHz with extremely high power
B 3 MHz to 30 MHz with low power
C 118 MHz to 137 MHz with low power

Level 2 · Page 36

95 **Why is the ground wave signal strength at a given point nearly constant?**

A The ionosphere does not change
B The earth's electrical properties are fairly constant
C The transmitter power is regulated

Level 2 · Page 36

96 **Where does the ground wave signal strength NOT stay nearly constant?**

A In areas close to the ocean
B In areas at high altitude
C In areas with distinct rainy and dry seasons

Level 2 · Page 36

97 **Why are high power, low frequency transmitters built near the ocean?**

A Sea air reduces ionospheric absorption
B Salt water gives superior surface wave conduction
C Water reflects the sky wave upward

Level 2 · Page 36

98 **How is some of the sky wave energy lost?**

A To particles in the atmospheric layers
B To eddy currents set up in the earth
C By reflection from the sea surface

Level 2 · Page 38

99 **Which is an advantage of sky wave propagation?**

A It is the same by day and by night
B It supports large distance propagation
C It needs only small antennas

Level 2 · Page 39

100 **Which is a disadvantage of sky wave propagation?**

A Long distances need large antennas
B Its atmospheric attenuation is high
C It only covers short distances

Level 2 · Page 39

101 **Which applications widely use sky wave propagation?**

A Mobile and satellite communications
B Marker beacons and glide slope
C Weather radar and radio altimeter

Level 2 · Page 39

102 **What does the ground wave induce on the surface of the earth?**

A A wave of half size, same polarity
B A wave of equal size, opposite polarity
C A wave of double size, same polarity

Level 3 · Page 37

103 **Where can strong sky wave signals be received?**

A Only directly below the transmitter
B Several hundred miles beyond the ground wave range
C Only inside the ground wave range

Level 3 · Page 38

104 **Why does sky wave transmission vary between day and night?**

A The transmitter power changes with daylight
B The ground conductivity changes with daylight
C The ionosphere is nearer at night and farther by day

Level 3 · Page 39

Coverage All text on pages 32–39 covered. Pages 33 and 35 are pictures only.
Conflict Page 34 says 'High frequencies (from HF and above) only travel in a direct line', but the same page and pages 32 and 39 say HF reflects from the ionosphere as a sky wave. Questions follow each statement as written.
Conflict Page 39 says sky wave propagation is 'widely used in mobile and satellite communications'. This does not fit the rest of the manual (satellite links are space waves). The question follows the LTT wording.
Conflict Page 39 says the ionosphere is 'near and far during night and day respectively'. The wording is unclear; the question follows it as written.

6 · Ionosphere and Ionospheric Effects

Pages 40-47 · 41 questions

HIGHLIGHTED

105 **Where is the ionosphere found?**

A About 40 to 50 miles above the earth
B About 4 to 5 miles above the earth
C About 400 to 500 miles above the earth

Level 1 · Page 40

106 **Where does the F layer extend?**

A From 50 miles up to 90 miles only
B From 40 miles up to 50 miles only
C From 90 miles to the ionosphere's upper limit

Level 1 · Page 41

107 **What makes the ionosphere appear to have layers?**

A The lines of the earth's magnetic field
B Temperature differences at different heights
C Densities of ionization at different heights

Level 2 · Page 40

108 **What effect does the D layer have on radio waves?**

A Little effect, except absorption of energy
B Strong refraction back to earth
C Total reflection of VHF signals

Level 2 · Page 41

109 **When is the D layer present?**

A Both day and night
B Only during the night
C Only during the day

Level 2 · Page 41

110 **When is the E layer at its maximum density?**

A Shortly after sunset
B At about noon local time
C During early afternoon

Level 2 · Page 41

111 **When is the F2 layer usually at its greatest density?**

A At about noon local time
B Shortly after sunset
C During early afternoon hours

Level 2 · Page 41

112 **What happens to the F1 and F2 layers shortly after sunset?**

A They recombine into a single F layer
B They separate into two parts
C They both disappear completely

Level 2 · Page 41

113 **What are erratic patches of ionized atmosphere at E layer heights called?**

A Critical frequency zones
B Sporadic-E ionisation
C D layer absorption

Level 2 · Page 41

114 **What effect can strong sporadic ionisation have?**

A Harm electronic transmissions
B Remove the D layer completely
C Double the critical frequency

Level 2 · Page 41

115 **What happens to waves when they reach the ionosphere?**

A All are reflected back to earth
B All are absorbed by the layers
C Some pass into space, others bend

Level 2 · Page 44

116 **What is the angle of the group 2 rays called?**

A The elevation limit
B The critical angle
C The skip angle

Level 2 · Page 44

117 **At what angle do the group 3 rays strike the ionosphere?**

A The largest angle that still returns to earth
B The smallest angle that still returns to earth
C An angle that is too nearly vertical

Level 2 · Page 44

118 **What is the critical frequency?**

A The frequency at which the D layer absorbs all energy
B The lowest frequency sent straight up that still returns
C The highest frequency sent straight up that still returns

Level 2 · Page 44

119 **How far can daylight transmissions reach using E layer refraction?**

A Up to 150 miles
B Up to 1500 miles
C Up to 15 000 miles

Level 3 · Page 41

120 **What can sporadic-E ionisation enable?**

A Lower ground wave attenuation at LF
B Removal of the skip zone at night
C Good VHF transmissions over unusual distances

Level 3 · Page 41

121 **What decides whether the ionosphere returns a wave to earth?**

A The angle, the frequency and the ion density
B The power, the antenna height and the polarization
C The ground conductivity, the season and the antenna type

Level 3 · Page 44

122 **Why are the group 1 rays not returned to earth?**

A Their frequency is too low
B Their angle is too nearly vertical
C Their angle is too close to the horizon

Level 3 · Page 44

123 **What happens to a ray that leaves the antenna at an angle greater than the critical angle?**

A It is absorbed by the D layer
B It returns at a shorter distance
C It penetrates the ionosphere

Level 3 · Page 44

124 **What happens to rays sent at a smaller angle than the group 3 rays?**

A They pass straight through the ionosphere
B They are refracted but do not return to earth
C They return to earth near the transmitter

Level 3 · Page 44

NOT HIGHLIGHTED

125 **What does the ionosphere have more of than the rest of the atmosphere?**

A Positive and negative ions
B Water vapour
C Neutral air molecules

Level 1 · Page 40

126 **Above what altitude is the air too thin for large-scale energy transfer?**

A 350 miles
B 50 miles
C 90 miles

Level 1 · Page 40

127 **At what altitude is the D layer?**

A Between 50 and 90 miles
B Above 90 miles
C Between 40 and 50 miles

Level 1 · Page 41

128 **At what altitudes is the E layer found?**

A Between 40 and 50 miles
B Between 50 and 90 miles
C Above 90 miles

Level 1 · Page 41

129 **At what altitude is the E layer densest?**

A About 70 miles
B About 150 miles
C About 45 miles

Level 1 · Page 41

130 **What raises atoms in the ionosphere to a high energy level?**

A The earth's magnetic field
B Solar ultraviolet and particle radiation
C Heat radiated from the earth

Level 2 · Page 40

131 **Which factors affect the number of ions in the ionosphere?**

A Transmitter power, antenna height and polarization
B Ground conductivity, rainfall and sea state
C Earth's rotation, its course round the sun and sunspots

Level 2 · Page 40

132 **Why are there only a few ions below about 40 miles altitude?**

A The air particles are far too sparse
B The ultraviolet radiation is strongest there
C The rate of return to a normal energy level is high

Level 2 · Page 40

133 **How does the D layer affect transmissions through daylight zones?**

A It changes their polarization
B It greatly reduces their field intensity
C It greatly increases their range

Level 2 · Page 41

134 **How does the F layer change between night and day?**

A One layer at night, two parts (F1, F2) by day
B It is present only by day
C Two parts at night, one layer by day

Level 2 · Page 41

135 **What does the critical frequency vary with?**

A Antenna type, polarization, power
B Ground conductivity and rainfall
C Locality, time, season and sunspots

Level 2 · Page 44

136 **What do nomograms and frequency tables predict?**

A The Maximum Usable Frequency for each hour and locality
B The ground wave range for each transmitter
C The skip distance for each antenna type

Level 2 · Page 45

137 **What is the skip zone?**

A The area beyond the point where the first sky wave returns
B The silent area between the end of the ground wave and the first sky wave
C The area that is covered by the ground wave alone

Level 2 · Page 45

138 **What is multiple-hop transmission?**

A Repeated reflection between ionosphere and ground
B Reception of the ground and sky wave together
C Transmission on several frequencies at once

Level 2 · Page 45

139 **What is the highest frequency the E layer can refract back to earth at midday?**

A 200 MHz
B 20 MHz
C 2 MHz

Level 3 · Page 41

140 **What happens to the initial angle as the frequency increases?**

A It increases
B It stays the same
C It decreases

Level 3 · Page 44

141 **What happens to waves at very high frequencies?**

A They are returned at any angle
B They return only when sent straight up
C They are not returned at any angle

Level 3 · Page 44

142 **When is the skip zone larger for 3 to 9 MHz?**

A At night
B At noon
C In early afternoon

Level 3 · Page 45

143 **How can skip effects be minimized at night?**

A By using higher frequencies
B By increasing the antenna height
C By using lower frequencies

Level 3 · Page 45

144 **What happens to the skip distance as the frequency decreases?**

A It decreases
B It stays the same
C It increases

Level 3 · Page 45

145 **What can cause fading in multiple-hop transmission?**

A The antenna changing its polarization
B Signals with different reflections superposing with different phases
C The D layer absorbing the ground wave

Level 3 · Page 45

Coverage All text on pages 40–47 covered. Pages 42–43 figures and page 47 picture text used.
Conflict Page 46 (figure): 'Skip Distance' is labelled from A to B in the figure only; the text never defines it. The question is marked as coming from a figure.

7 · Antenna Fundamentals, Fields, Polarization and Radiation Pattern

Pages 48-65 · 35 questions

● **HIGHLIGHTED**

● **146** **What does an antenna represent in a radio system?**

A The modulator of the transmitter
B The link between transmitter and receiver
C The power supply of the transmitter

Level 1 · Page 48

● **147** **What are antennas always supplied with?**

A AC, typically at high frequencies
B AC at mains frequency
C DC from the aircraft bus

Level 1 · Page 52

● **148** **What is the typical length of an antenna?**

A 2λ
B λ/4
C λ

Level 1 · Page 52

● **149** **What is the angle between the electrical and the magnetic field?**

A 180 degrees
B 90 degrees
C 0 degrees

Level 1 · Page 60

● **150** **What are the two most common types of antenna polarization?**

A Horizontal and vertical
B Circular and vertical
C Circular and horizontal

Level 1 · Page 60

● **151** **What is an antenna?**

A An amplifier for the received RF signal
B A filter that selects one frequency
C A wave transformer between line energy and free radiation

Level 2 · Page 48

● **152** **How is the current spread along a wire fed with voltage?**

A Greatest at the feed point, zero at the end
B Equal along the whole wire
C Zero at the feed point, greatest at the end

Level 2 · Page 50

153 **What shape does the current distribution along the wire have?**

A A rectangle
B A straight line
C Part of a sine

Level 2 · Page 50

154 **How are voltage and current related in a twin cable that acts like a coil?**

A The current leads the voltage
B The voltage leads the current
C They are in phase

Level 2 · Page 54

155 **Why does a twin cable hardly radiate?**

A The H and E fields are in phase
B The H and E fields stay directly near the line
C The H and E fields cancel completely

Level 2 · Page 54

156 **What do you get by spreading the lines of a twin cable apart?**

A A dipole antenna
B A coaxial cable
C A loop antenna

Level 2 · Page 56

157 **What is the field close to a transmitting antenna called?**

A The long distance field
B The radiation field
C The short distance field

Level 2 · Page 58

158 **What do displacement currents produce, according to Maxwell?**

A A direct current
B A magnetic field
C A static electric charge

Level 2 · Page 58

159 **What defines the polarization of a radio wave?**

A The direction of the electrical field
B The direction of the magnetic field
C The direction of propagation

Level 2 · Page 60

160 **How are aviation navigation and communication signals polarized?**

A Navigation vertical, communication horizontal
B Both are vertically polarized
C Navigation horizontal, communication vertical

Level 2 · Page 60

161 **Which antennas transmit circularly polarized waves?**

A Yagi antennas
B Helical antennas
C Loop antennas

Level 2 · Page 60

162 **What does circular polarization do?**

A It stays horizontal at all times
B It stays vertical at all times
C It continuously alters its polarization

Level 2 · Page 60

163 **In which direction does a λ/2 dipole radiate?**

A At 90° to its axis, virtually zero along its axis
B Along its axis, virtually zero at 90°
C Equally in all directions in space

Level 2 · Page 64

164 **After what distance has the short distance field practically gone?**

A 3 to 4 λ
B 0.1 λ
C 30 to 40 λ

Level 3 · Page 58

165 **What type of energy does the short distance field contain?**

A No energy
B Reactive energy
C Active energy

Level 3 · Page 58

166 **How are E and H related in the long distance field of an antenna?**

A In phase, transmitting active energy
B 180° out of phase, cancelling out
C 90° displaced, holding reactive energy

Level 3 · Page 58

167 **How does the Poynting vector change with distance and direction?**

A It is constant in all directions
B It falls in direct proportion to distance only
C It falls with distance squared and depends on direction

Level 3 · Page 62

NOT HIGHLIGHTED

168 **What does the radiation pattern of an antenna show?**

A The distribution of radiation in different directions
B The impedance over frequency
C The current along the antenna

Level 1 · Page 62

169 **Why can antenna theory be explained with a transmitting antenna only?**

A Transmitting antennas are always smaller
B Receiving antennas work at lower frequencies
C Both antenna kinds are electrically treated the same way

Level 2 · Page 48

170 **Why do transmitting and receiving antennas differ mechanically?**

A Because of different polarizations
B Because of varying high transmission power
C Because of different frequencies

Level 2 · Page 48

171 **Which criteria decide the choice of antenna?**

A Weight, cost, mounting and cable length
B Cable length, connector type and voltage
C Frequency range, bandwidth, propagation and polarization

Level 2 · Page 48

172 **Where is the voltage highest on a twin cable?**

A Midway along the wires
B At the end of the wires
C At the feed point

Level 2 · Page 54

173 **How close to the antenna is the short distance field found?**

A $r > 3–4\ \lambda$
B $r < 0.1\ \lambda$
C $r = 1\ \lambda$

Level 2 · Page 58

174 **What is the phase relation of H and E in the short distance field?**

A 90° displaced
B 180° displaced
C In phase

Level 2 · Page 58

175 **What shows the direction the radiated energy travels in?**

A The magnetic field H
B The electrical field E
C The Poynting vector S

Level 2 · Page 58

176 **How are home radio and TV signals polarized?**

A Both horizontal
B Radio vertical, TV horizontal
C Radio horizontal, TV vertical

Level 2 · Page 60

177 **What is a theoretical antenna that radiates equally in all directions called?**

A A Marconi antenna
B An isotropic radiator
C A λ/2 dipole

Level 2 · Page 62

178 **What shape does the dipole radiation have from a side view?**

A A figure eight
B A cardioid
C A circle

Level 2 · Page 64

179 **How does a dipole emit when seen from the top?**

A Equally in all directions
B Only forward and back
C Only along its axis

Level 2 · Page 64

180 **How is the Poynting vector oriented to the E and H fields?**

A Parallel to H
B At right angles to both
C Parallel to E

Level 3 · Page 58

Coverage All text on pages 48–65 covered. Page 49 figure is covered by the Yagi questions in T8. Pages 51, 53, 55, 57, 63 and 65 are figures.
Conflict Page 52: the text reads 'the typical length is /4' with the λ missing from the text layer; the figure on page 53 shows λ/4.

8 · Antenna Types

Pages 66-82 · 42 questions

● **HIGHLIGHTED**

● **181** **What does a Yagi antenna consist of?**

A A dipole plus one or more directors
B A loop plus one or more reflectors
C Two dipoles connected in parallel

Level 1 · Page 67

● **182** **What are Yagi directors also called?**

A Active elements
B Feed elements
C Parasitic elements

Level 1 · Page 67

● **183** **What is the rod antenna also referred to as?**

A Marconi antenna
B Yagi antenna
C Slot antenna

Level 1 · Page 69

● **184** **What is the loop antenna used for?**

A Receiving UHF signals
B Receiving long and medium wave
C Transmitting high frequencies

Level 1 · Page 71

● **185** **Which are examples of surface radiators?**

A The rod antenna and the loop antenna
B The Yagi antenna and the dipole
C The horn antenna and the lens antenna

Level 1 · Page 73

● **186** **What is a horn antenna?**

A A short rod with a ground plane
B A funnel-shaped waveguide
C A coiled wire loop

Level 1 · Page 73

● **187** **What is the impedance of free air?**

A 100π, about 314 Ω
B 150π, about 471 Ω
C 120π, about 377 Ω

Level 1 · Page 73

188

Which frequencies are horn antennas mainly used for?

A Below 30 MHz
B 1 GHz and above
C Between 3 MHz and 30 MHz

Level 1 · Page 73

189

Where is the radiating element of a reflector antenna installed?

A At the focal point of a parabolic reflector
B Behind the parabolic reflector
C At the rim of the parabolic reflector

Level 1 · Page 76

190

What is a delay lens antenna also known as?

A A metal plate lens
B A parabolic lens
C A dielectric lens

Level 1 · Page 79

191

What are the two kinds of dipole?

A Electrical dipole and magnetic dipole
B Yagi dipole and Marconi dipole
C Folded dipole and loop dipole

Level 1 · Page 82

192

How symmetric is the field of a dipole antenna?

A It is perfectly symmetric
B It is stronger on the lower half
C It is stronger on the upper half

Level 2 · Page 69

193

Why is the loop antenna well suited to direction finding?

A Because of its circular polarization
B Because of its very high gain
C Because of its directional characteristic

Level 2 · Page 71

194

What is the loop antenna NOT suitable for?

A As a direction-finding antenna
B As a transmitting antenna
C As a receiving antenna

Level 2 · Page 71

195

What does the funnel shape of a horn antenna act as?

A An amplifier for the waveguide signal
B A transformer between waveguide and free-air impedance
C A filter for unwanted frequencies

Level 2 · Page 73

196

What is needed for high antenna gain and strong bunching?

A Small apertures, wide flare angles, short lengths
B Large apertures, wide flare angles, short lengths
C Large apertures, small flare angles, long lengths

Level 2 · Page 76

197

How can low bunching in short horn antennas be overcome?

A By using lens and reflector antennas
B By using circular polarization
C By using a longer coaxial cable

Level 2 · Page 76

198

What does the focusing angle θ of a reflector antenna depend on?

A The polarization and the feed line length
B The frequency and the reflector diameter
C The power and the feed line length

Level 2 · Page 76

199

What is a slot antenna the equivalent of?

A The electrical equivalent of the loop
B A λ/4 rod on a ground plane
C The magnetic equivalent of the electrical dipole

Level 2 · Page 81

200

What gives a Yagi antenna its extra gain in the antenna direction?

A Superposition of the dipole and parasitic element fields
B The circular polarization of the dipole
C The greater length of the dipole

Level 3 · Page 67

201

How does a parasitic element's length decide its job?

A Shorter is a director, longer is a reflector
B Any length works as a director
C Shorter is a reflector, longer is a director

Level 3 · Page 67

202

How does the field of a λ/4 rod antenna relate to the field of a λ/2 dipole?

A It is identical to the dipole field
B It is twice the dipole field
C It corresponds to one half of the dipole field

Level 3 · Page 69

203

What does the induced voltage U0 of a loop antenna depend on?

A Loop area, number of windings and angle θ
B Loop shape, cable length and power
C Wire diameter, frequency and polarization

Level 3 · Page 71

204 **What increases the gain of a horn antenna?**

A Faster widening and a shorter funnel
B More gradual widening and a longer funnel
C A wider flare angle above 40°

Level 3 · Page 73

205 **What does a convex dielectric lens with εr > 1 achieve?**

A Centre beams radiate before edge beams
B Centre and edge beams radiate in phase
C Edge beams are absorbed by the lens

Level 3 · Page 76

206 **How do the beams leave a delay lens with the correct delay?**

A At different times and diverging
B At the same time and converging
C At the same time and in parallel

Level 3 · Page 76

NOT HIGHLIGHTED

207 **How is a slot antenna formed?**

A By a short rod on a ground plane
B By a coil of wire with several turns
C By a slot in a metal sheet or waveguide

Level 1 · Page 81

208 **What happens to the parasitic elements when the Yagi dipole transmits?**

A They are excited and transmit too
B They block the dipole's field
C They absorb all the energy

Level 2 · Page 67

209 **What creates the phase shift between the dipole and the parasitic elements?**

A The distance to the dipole
B The polarization of the dipole
C The length of the feed line

Level 2 · Page 67

210 **What happens to the Yagi antenna gain on the sides?**

A It is increased
B It is unchanged
C It is reduced

Level 2 · Page 67

211 **Where is the rod antenna used?**

A Where very high gain is needed
B Where circular polarization is needed
C Where the antenna must be as small as possible

Level 2 · Page 69

212 **Which part of the field induces the voltage in a loop antenna?**

A The electrical component
B The magnetic component
C The Poynting vector

Level 2 · Page 71

213 **What does the propagation velocity of a wave in a lens depend on?**

A The wave polarization
B The εr of the lens material
C The reflector diameter

Level 2 · Page 76

214 **How many beams does a dish antenna produce compared with a lens antenna?**

A Both of them give several beams
B The dish gives several beams, the lens one
C The dish gives one beam, the lens several

Level 2 · Page 78

215 **Which principle do lens antennas use?**

A The converging lens, to collimate energy into plane waves
B The parabolic mirror, to reflect the energy
C The diverging lens, to spread the energy

Level 2 · Page 79

216 **How long is a Yagi director compared with the λ/2 dipole?**

A λ/2 − 5 %
B λ/2 + 5 %
C λ/4

Level 3 · Page 68

217 **What happens to the field above a perfectly conductive plane placed in the middle of a dipole?**

A It disappears completely above the plane
B It stays the same, even without the lower half
C It doubles in strength above the plane

Level 3 · Page 69

218 **When is the maximum voltage induced in a loop antenna?**

A When the loop plane lies parallel to the ground
B When the loop plane is at 90° to the propagation direction
C When the loop plane lines up with the propagation direction

Level 3 · Page 71

219 **Which horn dimensions give good focusing?**

A Flare under 40°, length and width large compared to λ
B Flare of 90°, length and width equal to λ
C Flare over 40°, length and width small compared to λ

Level 3 · Page 73

220 **Which beams are delayed most by a delay lens?**

A All beams equally
B Beams that hit the lens edge
C Beams that hit the lens centre

Level 3 · Page 76

221 **Why does a wave undergo retardation in a delay lens?**

A The lens reflects part of the wave
B The lens medium increases the electrical path length
C The lens reduces the frequency

Level 3 · Page 79

222 **What polarization does a vertically installed slot antenna produce?**

A Circular
B Vertical
C Horizontal

Level 3 · Page 81

Coverage All text on pages 66–82 covered. Page 75 and 80 are pictures (lens simulations) with no testable text.
Conflict Page 78 says the lens antenna 'focuses RF waves thru Reflection'. A lens works by refraction (page 79 also describes it as a converging lens). No question was written on that line.

9 · Antenna Characteristics and Antenna Tuning

Pages 83-89 · 27 questions

● **HIGHLIGHTED**

● **223** **What does the lobe width indicate?**

A The bandwidth of an antenna
B The directivity of an antenna
C The impedance of an antenna

Level 1 · Page 83

● **224** **What is the usual reference antenna for antenna gain?**

A A λ/4 rod antenna
B A Yagi antenna
C An isotropic antenna

Level 1 · Page 83

● **225** **What unit is antenna gain normally given in?**

A Decibels
B Watts
C Ohms

Level 1 · Page 83

● **226** **What is the effect of concentrating the radiation in one direction?**

A Greater field strength there for the same power
B Lower field strength there for the same power
C No change in field strength

Level 2 · Page 83

● **227** **What is antenna gain G?**

A The power lost along the feed line, compared to the input
B Extra radiated power in a direction, compared to a reference antenna
C The antenna's physical length, compared to the wavelength

Level 2 · Page 83

● **228** **What is antenna gain measured against?**

A The receiver noise
B The feed line loss
C A reference source

Level 2 · Page 85

● **229** **How is an antenna tuned mechanically?**

A Its length is adjusted to the wavelength
B Its feed line is shortened
C Its polarization is changed

Level 2 · Page 86

230 **How can a short antenna still be brought to resonance?**

A By lengthening the feed line
B With an L or T shaped tip or an umbrella
C By using a loop instead

Level 2 · Page 86

231 **How is an antenna's length adjusted when it is tuned?**

A Only electrically to the polarization
B Only mechanically to the power
C Mechanically or electrically to the frequency

Level 2 · Page 88

232 **What is the diagram of an antenna's resistance and reactance over frequency called?**

A The resonance curve
B The locus
C The radiation pattern

Level 2 · Page 88

233 **What must the reactance be to tune the antenna to the desired frequency?**

A Equal to R
B Zero
C Maximum

Level 2 · Page 88

234 **What tunes an HF antenna over its frequency range?**

A The HF coupler
B The power amplifier
C The antenna feeder

Level 2 · Page 89

235 **What does the end capacitance do to the antenna?**

A Leaves its resonant frequency and length unchanged
B Raises its resonant frequency, electrically shortening it
C Lowers its resonant frequency, electrically lengthening it

Level 3 · Page 86

NOT HIGHLIGHTED

236 **What angle does the lobe width describe?**

A Angle where the radiation is zero
B Apex angle where power falls to half
C Angle of maximum radiated power

Level 2 · Page 83

237 **Which dB point is the lobe width equivalent to?**

A The 6 dB point
B The 3 dB point
C The 10 dB point

Level 2 · Page 83

238 **What is the effect of using a directional antenna?**

A The same as increasing transmitter power
B The same as using a longer feed line
C The same as reducing the frequency

Level 2 · Page 83

239 **Which antennas are often used as reference antennas in practice?**

A λ/4 rods
B λ/2 dipoles
C Loop antennas

Level 2 · Page 83

240 **What is the antenna boresight?**

A The axis of maximum gain
B The axis of the feed line
C The axis of zero radiation

Level 2 · Page 85

241 **What is the gain of an isotropic antenna?**

A 0 dBi
B 3 dBi
C 1.64 dBi

Level 2 · Page 85

242 **What does the locus show on its X-axis and Y-axis?**

A Frequency on X, gain on Y
B Reactance on X, resistance on Y
C Resistance on X, reactance on Y

Level 2 · Page 88

243 **What is the gain of a λ/2 dipole compared to an isotropic antenna?**

A 3.0
B 1.0
C 1.64

Level 3 · Page 83

244 **Why must a long wave antenna be shorter than λ/4?**

A Long wave wavelengths can be several kilometres
B Long wave antennas need high gain
C Long wave needs circular polarization

Level 3 · Page 86

245 **How is the end capacitance CE connected to the antenna capacitance CA?**

A In parallel, giving a larger total capacitance
B In series, giving a smaller total capacitance
C It replaces CA completely

Level 3 · Page 86

246 **How is an antenna that is too short for f1 tuned?**

A With an inductor in parallel
B With a capacitor in series
C With an inductor in series

Level 3 · Page 88

247 **How is an antenna that is too long for f2 tuned?**

A With a capacitor in parallel
B With a capacitor in series
C With an inductor in series

Level 3 · Page 88

248 **How is an antenna that is too short for f3 tuned?**

A With a capacitor in series
B With a capacitor in parallel
C With an inductor in parallel

Level 3 · Page 88

249 **How is an antenna that is too long for f4 tuned?**

A With a capacitor in parallel
B With an inductor in series
C With an inductor in parallel

Level 3 · Page 88

Coverage All text on pages 83–89 covered. Page 84 figure (lobe width, −3 dB) covered by the lobe width questions. Page 87 is a figure.

10 · RF Lines, Matching and Standing Waves

Pages 90-103 · 40 questions

● **HIGHLIGHTED**

● **250** **What are RF lines used for?**

A Transferring RF energy with as little loss as possible
B Generating the basic radio frequency
C Converting RF energy into free radiation

Level 1 · Page 90

● **251** **What else are RF lines used for?**

A Polarization, focusing and filtering
B Modulation, detection and amplification
C Resonant circuits, resistance transformation and matching

Level 1 · Page 90

● **252** **What type of line is a coaxial cable?**

A An unshielded line
B A symmetric line
C An asymmetric line

Level 1 · Page 90

● **253** **What impedance do typical symmetric twin cables start at?**

A About 377 Ω
B About 150 Ω
C About 50 Ω

Level 1 · Page 94

● **254** **What is the impedance matching principle called?**

A Power transfer theorem or Jacobi's Law
B Kirchhoff's first law
C Thomson's oscillation formula

Level 1 · Page 96

● **255** **How are RF lines grouped, electrically and physically?**

A Electrically symmetric or asymmetric, physically shielded or not
B Electrically matched or not, physically open or shorted
C Electrically shielded or not, physically symmetric or asymmetric

Level 2 · Page 90

● **256** **What are the features of a symmetric cable?**

A One line grounded, one line at RF potential
B Equal capacity to ground, equal and opposite RF signals
C A shield around a concentric inner line

Level 2 · Page 90

257 **What are the features of an asymmetric cable?**

A Both lines at equal RF potential
B One line at RF potential, the other grounded
C Both lines with equal capacity to ground

Level 2 · Page 90

258 **What determines the electrical data of RF lines?**

A The antenna type and polarization
B The applied frequency and power
C The physical construction and the dielectric

Level 2 · Page 92

259 **What sets the ratio of current to voltage (ix to ux) on a line?**

A The series and parallel resistance
B The line inductance L' and capacity C'
C The length of the line

Level 2 · Page 92

260 **How is the impedance of a cable generally calculated?**

A $Z0 = L \times C$
B $Z0 = \sqrt{(L/C)}$
C $Z0 = \sqrt{(C/L)}$

Level 2 · Page 94

261 **What are the disadvantages of coax cables with Teflon or polyethylene dielectric?**

A High price and difficult handling
B Lower dampening, higher voltage rating, higher power
C Higher dampening, lower voltage rating, lower power

Level 2 · Page 94

262 **What is needed for maximum energy transfer from transmitter to antenna?**

A The line must be as long as possible
B The antenna must be longer than λ
C The transmitter and antenna impedances must match

Level 2 · Page 96

263 **What does a mismatched antenna create on an RF line?**

A A standing wave
B An open line
C A matched line

Level 2 · Page 98

264 **What does the standing wave ratio indicate?**

A How far the antenna and line impedances differ
B How much power the antenna actually radiates
C How long the RF line between them is

Level 2 · Page 98

265 **What does an SWR of 1 mean?**

A Shorted line, maximum reflection
B Open line, no power transfer
C Identical impedances, maximum power transfer

Level 2 · Page 98

266 **What is the SWR of an open or shorted line?**

A 1
B Infinite
C 0

Level 2 · Page 98

267 **Which two kinds of reflection can occur on an RF line?**

A Opposite phase and in-phase reflection
B Hard and soft absorption
C Ground and sky reflection

Level 2 · Page 99

268 **What appears on a line with a shorted or open end?**

A No current and no voltage
B A perfectly matched load
C Large standing waves for current and voltage

Level 2 · Page 102

269 **Why does an asymmetric cable have no C20 capacity to ground?**

A Ground itself is the conductor
B The line is shielded
C The line is too short

Level 3 · Page 90

270 **Which capacities does the inner wire of a coaxial cable have?**

A C10 to ground, no C12 to the shield
B Equal C10 and C20 to ground
C C12 against the shield, no C10 to ground

Level 3 · Page 90

271 **What creates a hard reflection on an RF line?**

A A small impedance deviation
B A short or an open line
C A perfectly matched load

Level 3 · Page 98

272 **What happens at the end of an open RF line?**

A Current and voltage are both zero
B No current flows, the voltage is at maximum
C Maximum current flows, the voltage is zero

Level 3 · Page 100

273 **What happens at the end of a shorted RF line?**

A Current and voltage are both at maximum
B No current flows, the voltage is at maximum
C Maximum current flows, the voltage is zero

Level 3 · Page 100

274 **What does a shorted λ/4 line act like?**

A A parallel resonant circuit
B A series resonant circuit
C A pure inductor

Level 3 · Page 102

275 **What does an open λ/4 line act like?**

A A series resonant circuit
B A parallel resonant circuit
C A pure capacitor

Level 3 · Page 102

NOT HIGHLIGHTED

276 **What impedances are common for coaxial cables?**

A 377 Ω
B 150 Ω to 300 Ω
C 50 Ω to 75 Ω

Level 1 · Page 94

277 **What is the standard impedance of coaxial power cables in modern installations?**

A 150 Ω
B 50 Ω
C 75 Ω

Level 1 · Page 94

278 **Which values does every line have per unit length?**

A L', C', series and parallel resistance
B Only the line inductance L'
C Only the line capacity C'

Level 2 · Page 92

279 **What determines the line impedance Z0?**

A The amount of power fed into the line
B The voltage/current ratio of an infinitely long line
C The physical length of the line in use

Level 2 · Page 94

280 **What load resistance gives a matched RF line?**

A RL = 2 Z0
B RL = Z0
C RL = 0

Level 2 · Page 97

281 **What happens when the reflected wave meets the initial wave?**

A They both stop at that point
B They pass with no effect at all
C They superpose, amplify or cancel

Level 2 · Page 98

282 **What does a greater SWR indicate?**

A Greater impedance deviation
B Smaller impedance deviation
C A better match of impedances

Level 2 · Page 98

283 **How is the SWR calculated from voltage?**

A E maximum × E minimum
B E maximum ÷ E minimum
C E minimum ÷ E maximum

Level 2 · Page 98

284 **How is the SWR worked out from the load and line impedances?**

A Zload − Z0, whichever is greater
B Zload × Z0, whichever is greater
C Zload ÷ Z0 or Z0 ÷ Zload, whichever is greater

Level 2 · Page 98

285 **What damage can a very high voltage on an RF line cause?**

A Burned spots from heating
B Loss of polarization
C Arcing and sparking in the dielectric

Level 2 · Page 100

286 **What damage can a very high current on an RF line cause?**

A Arcing in the dielectric
B Heated spots, burns or small fires
C A change of impedance only

Level 2 · Page 100

287 **How far apart are the in-phase peaks on a hard-reflecting line?**

A Half a wavelength
B A quarter wavelength
C One full wavelength

Level 3 · Page 100

288 **What does an open λ/2 line act like?**

A A series resonant circuit
B A parallel resonant circuit
C A matched load

Level 3 · Page 102

289 **What do current and voltage show at the input of a line acting as a parallel resonant circuit?**

A A current loop and a voltage loop
B A current loop and a voltage node
C A current node and a voltage loop

Level 3 · Page 102

Coverage All text on pages 90–103 covered. Pages 91, 93, 95, 97, 99, 101, 103 are figures; labels used where testable.

11 · Resonant Circuits

Pages 105-117 · 35 questions

● **HIGHLIGHTED**

● **290** **What does a mechanical oscillator consist of?**

A A mass and a return force
B A coil and a capacitor
C A spring and a resistor only

Level 1 · Page 105

● **291** **What do electric oscillator systems consist of?**

A Inductors and capacitors
B Resistors and diodes
C Transformers and resistors

Level 1 · Page 105

● **292** **What are resonant circuits used for?**

A Generation of a steady DC voltage
B Selective amplification, muting, transformation
C Rectification of AC into DC voltage

Level 1 · Page 116

● **293** **How is energy stored in an LC oscillator?**

A Both of them store it as heat in the resistor
B The inductor in a magnetic field, the capacitor as voltage
C The inductor as voltage, the capacitor in a magnetic field

Level 2 · Page 105

● **294** **What happens in a series resonant circuit at the resonant frequency f0?**

A UL and UC cancel, so a high current flows, limited by RS
B XL is greater than XC, so the circuit is inductive
C iL and iC cancel, so only the current iR flows

Level 2 · Page 109

● **295** **What is the bandwidth B of a resonant circuit?**

A The frequency range where the curve drops to a set value each side of the peak
B The frequency at the peak of the resonance curve
C The total range of frequencies the circuit can pass

Level 2 · Page 114

● **296** **What determines the bandwidth?**

A The applied voltage
B The resistor value only
C The quality factor Q

Level 2 · Page 114

297 **What is selectivity?**

A Suppressing frequencies other than the resonant one
B Raising the resonant frequency
C Passing the widest band of frequencies

Level 2 · Page 114

298 **How large can the voltages across L and C be in a series circuit at f0?**

A Always lower than the applied voltage
B Much higher than the applied voltage
C Always equal to the applied voltage

Level 3 · Page 109

299 **What happens in a parallel resonant circuit at the resonant frequency f0?**

A iL is greater, so the circuit is inductive
B UL and UC cancel, so a high current flows
C iL and iC cancel, so only iR flows

Level 3 · Page 111

300 **How is a strong unwanted local transmitter suppressed?**

A A series circuit tuned to it, used as the ground leg
B A series circuit tuned to the wanted signal, in line
C A parallel circuit tuned to it, used as the ground leg

Level 3 · Page 116

301 **How is one desired frequency boosted at the receiver input?**

A A series circuit tuned to it, used as the ground leg
B A parallel circuit tuned to the unwanted signal
C A parallel circuit tuned to it, used as the ground leg

Level 3 · Page 116

NOT HIGHLIGHTED

302 **Why does energy travel back and forth between inductor and capacitor?**

A They are fed from a DC source
B They both behave as pure resistors
C They create a time delay between voltage and current

Level 2 · Page 105

303 **What determines the frequency of a mechanical pendulum?**

A The mass and the length l
B The external force
C The amplitude only

Level 2 · Page 107

304 **What happens to the sum of potential and kinetic energy in a pendulum without losses?**

A It increases
B It stays constant
C It decreases

Level 2 · Page 107

305 **How is energy introduced into an LC oscillator?**

A A magnet is moved near the coil
B A resistor is placed in series
C An external voltage charges the capacitor

Level 2 · Page 107

306 **What is Thomson's oscillation formula?**

A f0 = √(L/C) / 2π
B f0 = 2π√(LC)
C f0 = 1 / (2π√(LC))

Level 2 · Page 107

307 **Which pendulum energy matches the magnetic field energy of the oscillator?**

A Potential energy
B Kinetic energy
C Heat energy

Level 2 · Page 108

308 **What does the geometric sum of UL, UC and UR equal in a series circuit?**

A The applied voltage U
B The current i
C Zero at all times

Level 2 · Page 109

309 **When is the impedance Z of a series resonant circuit lowest?**

A Below f0
B Above f0
C At f0

Level 2 · Page 109

310 **What is the phase angle between U and i in a series circuit at resonance?**

A Zero degrees
B +90 degrees
C −90 degrees

Level 2 · Page 109

311 **What does the total current equal in a parallel circuit?**

A Zero at all frequencies
B The geometric sum of the branch currents
C The current in the capacitor only

Level 2 · Page 111

312 **What does the impedance Z of a parallel circuit equal at resonance?**

A Zero
B RP
C XL + XC

Level 2 · Page 111

313 **How does the impedance of a resistor change with frequency?**

A It stays constant
B It rises in proportion
C It falls non-linearly

Level 2 · Page 113

314 **How does the impedance of an inductor change with frequency?**

A It stays constant
B It falls non-linearly
C It rises in proportion

Level 2 · Page 113

315 **How does the impedance of a capacitor change as frequency increases?**

A It decreases, not linearly
B It stays constant
C It increases in proportion

Level 2 · Page 113

316 **What is the frequency at which XL = XC called?**

A The critical frequency
B The cut-off frequency
C The resonant frequency fR

Level 2 · Page 113

317 **At what level is the normal bandwidth measured?**

A −10 dB (0.1)
B −3 dB (0.707)
C −6 dB (0.5)

Level 2 · Page 114

318 **How is the bandwidth worked out from the quality factor Q?**

A B = f0 / Q
B B = Q / f0
C B = f0 × Q

Level 2 · Page 114

319 **When does the magnetic field in the LC oscillator reach maximum strength?**

A When the switch is open
B When the voltage is maximum
C When the voltage is zero

Level 3 · Page 107

320 **What charges the capacitor with reversed polarity in an LC oscillator?**

A The losses in the resistor
B Voltage induced by the coil's field
C The external voltage source

Level 3 · Page 107

321 **How does a series resonant circuit behave below and above f0?**

A Inductive below f0, capacitive above f0
B Resistive at all frequencies
C Capacitive below f0, inductive above f0

Level 3 · Page 109

322 **How does a parallel resonant circuit behave below and above f0?**

A Inductive below f0, capacitive above f0
B Capacitive below f0, inductive above f0
C Resistive at all frequencies

Level 3 · Page 111

323 **What is the trade-off of a high selectivity?**

A It comes at the cost of gain
B It comes at the cost of bandwidth
C It comes at the cost of voltage

Level 3 · Page 114

324 **What does a large Q factor do to the resonance curve?**

A Makes it gradual
B Makes it sharp
C Makes it flat

Level 3 · Page 115

Coverage All text on pages 105–117 covered. Pages 106, 108, 110, 112, 115 and 117 are figures; page 108 and 115 labels used.
Conflict Page 105: 'displaces' is a typo for 'displaced'. Page 107 spells the formula name 'Thompsons'; questions use 'Thomson'.

Answer Key

All 324 questions · Key balance A 108 · B 108 · C 108

1 · Radio Technology Basics

#	Ans	Why (LTT wording)	Page
1	**B**	Both are basically made of power supplies, amplifiers and oscillators.	6
2	**B**	A basic radio communication system may consist of only a transmitter and a receiver.	6
3	**A**	The stage required to place the audio intelligence on the RF signal is the modulator.	6
4	**A**	Sonic waves are converted to electrical signals by microphones.	6
5	**C**	Loudspeakers convert electrical signals to sonic waves.	6
6	**A**	They are connected by the medium through which the electromagnetic waves travel.	6
7	**B**	No medium is necessary for the propagation of electromagnetic waves.	6
8	**C**	The transmitter comprises an oscillator, RF amplifiers and the modulator stage.	6
9	**A**	The oscillator generates a basic radio frequency.	6
10	**B**	The receiving antenna turns the energy into alternating RF currents.	6
11	**C**	The receiver turns the current variations back into the intelligence (the message).	6

2 · AC Properties, Frequency and Angular Frequency

#	Ans	Why (LTT wording)	Page
12	**A**	In communications engineering both AC as well as DC is used.	8
13	**C**	The time for one cycle to complete is called cycle time T.	8
14	**A**	The symbol for period is T.	8
15	**B**	The period is measured in seconds (s).	8
16	**C**	The frequency of AC is the number of complete cycles, i.e. oscillations, in one second.	10
17	**A**	1 cps = Hz.	10
18	**B**	$\omega = 2\pi \times f$.	12
19	**C**	The current I varies in direction (polarity) and in quantity over a period of time t.	8
20	**B**	The change of polarity and current repeats periodically, and so does the voltage.	8
21	**C**	One cycle is a positive half-wave plus a negative half-wave.	9
22	**B**	Frequency is also defined as the reciprocal of the cycle time T: $f = 1/T$.	10
23	**A**	The rotational speed of the tip depends on the RPM and the diameter of the generator.	12
24	**B**	The length is the voltage amplitude; the rotation speed is set by the frequency.	12
25	**C**	Angular velocity is a vector quantity, while angular frequency is a scalar quantity.	13
26	**A**	AC is the type of current used in the mains supply in the field of power engineering.	8

#	Ans	Why (LTT wording)	Page
27	**B**	Every generator containing rotating parts will create a sinusoidal AC.	8
28	**C**	The frequency is measured in hertz (Hz).	10
29	**C**	It rises from 0 to the positive peak and falls back to 0.	8
30	**B**	The higher the frequency, the shorter the cycle time.	10

3 · Harmonic, Triangular and Sawtooth Waves

#	Ans	Why (LTT wording)	Page
31	**C**	The sine wave at the same frequency is the basic oscillation, or 1st harmonic.	16
32	**A**	Multiples of the basic frequency are further harmonic waves.	16
33	**C**	Adding the 1st, 3rd and 5th harmonics gives an almost rectangular wave.	16
34	**A**	Triangular waves use the odd harmonics, like rectangular waves.	19
35	**B**	For sawtooth waves all harmonics are required, the odd ones and the even ones.	19
36	**C**	Rectangular waves can be resolved to an infinite number of uneven harmonics.	16
37	**B**	The shape of the non-sinusoidal wave decides its harmonics and their amplitudes.	16
38	**A**	Divide the 1st harmonic's amplitude by the harmonic number.	16
39	**C**	For triangular waves, amplitude falls with the square of the harmonic number.	19
40	**A**	For sawtooth waves the amplitude decreases with the number of the harmonic.	19

4 · Frequency Bands, Wavelength and Aviation Frequencies

#	Ans	Why (LTT wording)	Page
41	**B**	The three most important frequency bands are HF, VHF and UHF.	21
42	**C**	HF high frequency: 3 MHz – 30 MHz.	21
43	**B**	VHF very high frequency: 30 MHz – 300 MHz.	21
44	**A**	UHF ultra-high frequency: 300 MHz – 3000 MHz.	21
45	**B**	Data communication, radio broadcast and satellite broadcast are communication uses.	22
46	**A**	In free space (vacuum) it would propagate with the speed of light c.	23
47	**A**	$\lambda = c \times T = c / f$ [m].	23
48	**C**	In a vacuum the speed of light c would be 299 792 km/s or approximately 300 000 km/s.	23
49	**B**	Radio frequencies up to 300 GHz are divided into decadic ranges.	27
50	**B**	Each decade is designated as a frequency band.	27
51	**C**	The frequencies used for radar technology are divided into special radar bands.	27
52	**C**	VHF COM uses 118–137 MHz.	28
53	**A**	HF COM uses 2–30 MHz.	28
54	**B**	UHF COM uses 225–400 MHz.	28
55	**A**	The ADF uses 200–1750 kHz.	28
56	**C**	SatCom uses 1.5 GHz.	28

#	Ans	Why (LTT wording)	Page
57	**B**	Weather radar uses 9.375 GHz.	28
58	**A**	VLF navigation uses 10–24 kHz.	28
59	**C**	The DME identification tone is 1350 Hz.	28
60	**A**	VOR, ILS and ADF use a 1020 Hz identification tone.	28
61	**A**	Voice covers 300 Hz to 3 kHz.	28
62	**C**	The marker uses 400, 1300 and 3000 Hz tones.	28
63	**A**	The propagation speed depends on the medium used.	23
64	**B**	The higher the density of the medium the lower the propagation speed will be.	23
65	**B**	IF = Intermediate Frequency.	27
66	**C**	The X band covers 5200–10 900 MHz.	27
67	**B**	The L band covers 390–1550 MHz.	27
68	**A**	The localizer uses 108–112 MHz.	28
69	**C**	The VOR uses 112–118 MHz.	28
70	**B**	The glide slope uses 329.3–399.95 MHz.	28
71	**A**	The marker beacon uses 75 MHz.	28
72	**A**	The DME uses 962–1213 MHz.	28
73	**B**	ATC uses 1030–1090 MHz.	28
74	**C**	The LRRA (radio altimeter) uses 4.25–4.35 GHz.	28
75	**C**	General rule of thumb is to multiply by 10 (3 MHz × 10 = 30).	21
76	**B**	MF is listed as ground wave and sky wave.	30

5 · Radio Wave Propagation: Ground Wave and Sky Wave

#	Ans	Why (LTT wording)	Page
77	**C**	Radio waves travel at approximately the speed of light.	32
78	**B**	Part of the wave moves outward in contact with the ground to form the GROUND WAVE.	32
79	**A**	The rest of the wave moves upward and outward to form the SKY WAVE.	32
80	**C**	Wave propagation defines the way in which waves are transmitted from an end to another.	34
81	**B**	The ground wave is responsible for most of the daytime broadcast reception.	36
82	**C**	Short range at HF with low power, and long range at LF with very high power.	32
83	**A**	The sky wave is used for long-range high frequency daylight communication.	32
84	**A**	High frequencies (from HF and above) only travel in a direct line. (See conflict note.)	34
85	**C**	Space waves: VHF and higher travel in a Line of Sight (LOS).	34
86	**C**	Ground waves: MF and lower follow the curvature of the earth.	34
87	**A**	The conductivity of salt water is 5,000 times as great as that of dry soil.	36
88	**A**	Ground wave usually works up to about 2 MHz.	37

#	Ans	Why (LTT wording)	Page
89	**B**	The permissible frequency range for sky wave propagation lies between 3 MHz and 30 MHz.	39
90	**A**	It looks like a huge doughnut lying on the ground, with the antenna in the middle.	32
91	**B**	At night the sky wave gives long-range contacts at somewhat lower frequencies.	32
92	**C**	It sets up eddy currents in the earth, which take energy from the wave.	36
93	**C**	Higher frequency rapidly increases attenuation, so ground waves suit low frequencies.	36
94	**A**	Shore-base transmitters use frequencies between 18 and 300 kHz with extremely high power.	36
95	**B**	The earth's electrical properties along the path stay fairly constant.	36
96	**C**	Where there are rainy and dry seasons, moisture changes the soil's conductivity.	36
97	**B**	Because of the superiority of surface wave conduction by salt water.	36
98	**A**	Some energy is lost in dissipation to particles of the atmospheric layers.	38
99	**B**	Sky wave supports propagation over large distances.	39
100	**A**	Disadvantage: long-distance propagation requires large-sized antennas.	39
101	**A**	The LTT lists mobile and satellite communications. (See conflict note.)	39
102	**B**	Oscillations induce a wave of equal magnitude but opposite polarity on the surface.	37
103	**B**	Several hundred miles beyond the ground wave range, where the sky wave returns.	38
104	**C**	The LTT says the ionosphere is nearer at night and farther by day. (See conflict note.)	39

6 · Ionosphere and Ionospheric Effects

#	Ans	Why (LTT wording)	Page
105	**A**	The ionosphere starts about 40 to 50 miles above the earth.	40
106	**C**	The F layer runs from about 90 miles up to the top of the ionosphere.	41
107	**C**	Densities of ionization at different heights make the ionosphere appear to have layers.	40
108	**A**	Its ionization is low, so it only absorbs some energy.	41
109	**C**	The "D" layer is present only during the day.	41
110	**B**	The maximum density of the "E" layer appears at about noon local time.	41
111	**C**	As a rule, the F2 layer is at its greatest density during early afternoon hours.	41
112	**A**	Shortly after sunset, the F1 and F2 layers recombine into a single F layer.	41
113	**B**	These patches are referred to as Sporadic-E ionisation.	41
114	**A**	Strong sporadic ionisation can harm electronic transmissions.	41
115	**C**	Some pass right through into space; others penetrate but bend.	44
116	**B**	The angle made by the group 2 rays is called the CRITICAL ANGLE for that frequency.	44
117	**B**	Group 3 rays strike at the smallest angle that still returns to earth.	44

#	Ans	Why (LTT wording)	Page
118	**C**	It is the highest frequency sent straight up that still comes back to earth.	44
119	**B**	E layer refraction helps daylight transmissions up to 1500 miles.	41
120	**C**	Sporadic-E can allow good VHF transmissions over unusually long distances.	41
121	**A**	It depends on the angle, the frequency and the ion density.	44
122	**B**	Group 1 rays strike too nearly vertical, so they pass through and are lost.	44
123	**C**	Any ray above the critical angle penetrates the ionosphere.	44
124	**B**	At any smaller angle the rays will be refracted but will not return to the earth.	44
125	**A**	It contains a much higher number of positive and negative ions.	40
126	**A**	At altitudes above 350 miles the particles of air are far too sparse.	40
127	**C**	The ionized atmosphere between 40 and 50 miles is called the "D" layer.	41
128	**B**	The band of atmosphere between 50 and 90 miles contains the "E" layer.	41
129	**A**	It is a well defined band with greatest density at an altitude of about 70 miles.	41
130	**B**	Solar ultraviolet and particle radiation raise the atoms' energy.	40
131	**C**	The earth's rotation, its yearly path around the sun, and sunspots.	40
132	**C**	Below 40 miles, ions return to their normal energy level quickly.	40
133	**B**	The D layer greatly reduces the field strength of signals passing through it by day.	41
134	**A**	At night only one F layer is present; during the day it often separates into F1 and F2.	41
135	**C**	It varies with locality, time of day, season and the sunspot cycle.	44
136	**A**	They predict the Maximum Usable Frequency (MUF) for each hour and place.	45
137	**B**	No signals are heard between the end of the ground wave and the first returning sky wave.	45
138	**A**	The signal bounces between the ionosphere and the ground more than once.	45
139	**B**	At midday the E layer can sometimes return frequencies up to 20 MHz.	41
140	**C**	As the frequency increases, the initial angle decreases.	44
141	**C**	At high enough frequencies, waves are not returned at any angle.	44
142	**A**	For 3 to 9 MHz the skip zone is larger at night than by day.	45
143	**C**	Skip effects can be minimized by utilizing lower frequencies for nighttime communication.	45
144	**A**	As a general rule, as the frequency decreases, the skip distance decreases.	45
145	**B**	Signals with different numbers of hops arrive with different phases and interfere.	45

7 · Antenna Fundamentals, Fields, Polarization and Radiation Pattern

#	Ans	Why (LTT wording)	Page
146	**B**	It therefore represents the link between transmitter and receiver.	48
147	**A**	Antennas are always supplied with an AC, typically at high frequencies.	52
148	**B**	With antennas the typical length is λ/4.	52
149	**B**	The electrical and magnetic fields are at 90° to each other.	60

#	Ans	Why (LTT wording)	Page
150	**A**	The two most common polarizations used for antennas are horizontal and vertical.	60
151	**C**	An antenna turns energy on a line into free radiation (or back again).	48
152	**A**	The current is greatest at the feed point; it is zero at the end of the wire.	50
153	**C**	The current along the wire has the shape of part of a sine.	50
154	**B**	A coil shifts them by 90°, with the voltage leading the current.	54
155	**B**	In a twin cable the H and E fields stay right next to the line.	54
156	**A**	Spreading the two lines apart gives a dipole antenna.	56
157	**C**	This field is known as the antenna's short distance field.	58
158	**B**	Through an idea from Maxwell these traverse currents also produce a magnetic field.	58
159	**A**	The polarization of a radio wave is laid down as the direction of the electrical field.	60
160	**C**	In aviation: navigation is horizontal polarized; communication is vertical polarized.	60
161	**B**	Circular polarized waves are transmitted with helical antennas.	60
162	**C**	Circular polarization continuously alters its polarization.	60
163	**A**	A λ/2 dipole radiates at 90° to its axis, almost nothing along it.	64
164	**A**	The short distance field fades fast and is practically gone after 3–4 λ.	58
165	**B**	The short distance field contains therefore reactive energy.	58
166	**A**	Beyond 3–4 λ, E and H are in phase and active energy is sent.	58
167	**C**	It falls with the square of the distance and depends on direction.	62
168	**A**	It shows how the radiation is spread in different directions.	62
169	**C**	Transmission and reception antennae can electrically be dealt with the same way.	48
170	**B**	Mechanical differences are necessary due to varying high transmission power.	48
171	**C**	Important criteria: frequency range, band width, propagation characteristic, polarization.	48
172	**B**	The voltage is highest at the end of the wires but lowest at the feed point.	54
173	**B**	The short distance field is within 0.1 λ of the antenna.	58
174	**A**	In the short distance field H and E fields are 90° displaced.	58
175	**C**	The propagation direction of the radiated energy is given as the Poynting vector S.	58
176	**C**	Home radio is horizontally polarized; TV is vertically polarized.	60
177	**B**	A theoretical antenna radiating equally in all directions is an isotropic radiator.	62
178	**A**	From a side view, the dipole emits with the shape of an eight.	64
179	**A**	With a top view, the dipole emits in any direction with the same intensity.	64
180	**B**	S is at right angles to the direction of the E and H fields (S = E × H).	58

8 · Antenna Types

#	Ans	Why (LTT wording)	Page
181	**A**	A Yagi antenna consists of a dipole plus one or more directors.	67

#	Ans	Why (LTT wording)	Page
182	**C**	These directors often are referred to as parasitic elements.	67
183	**A**	The rod antenna is also referred to as Marconi-Antenna.	69
184	**B**	It is used as a receiver antenna for long and medium wave reception.	71
185	**C**	Examples of surface radiators are the Horn Antenna and the Lens Antenna.	73
186	**B**	A horn antenna is in effect a funnel shaped waveguide.	73
187	**C**	Free air impedance is 120π, approximately equal to 377 Ω.	73
188	**B**	Horn antennas are used mainly for frequencies of 1 GHz and above.	73
189	**A**	The radiating element sits at the focal point of the parabolic reflector.	76
190	**C**	These are sometimes known as a dielectric lens.	79
191	**A**	The slot antenna is compared as the magnetic dipole to the electrical dipole.	82
192	**A**	The field of a dipole antenna is perfectly symmetric.	69
193	**C**	Its directional characteristic makes it good for direction finding.	71
194	**B**	The loop antenna is not suitable for transmitting.	71
195	**B**	The funnel matches the waveguide impedance to the impedance of free air.	73
196	**C**	Large apertures, small flare angles and long overall lengths.	76
197	**A**	Lens and reflector antennas correct the phase and improve bunching.	76
198	**B**	The focusing angle θ depends upon the applied frequency and the reflector diameter.	76
199	**C**	The slot antenna is the magnetic equivalent to the electrical dipole.	81
200	**A**	The dipole and parasitic element fields add up (superimpose) in that direction.	67
201	**A**	Shorter than the dipole: director. Longer than the dipole: reflector.	67
202	**C**	A λ/4 rod's field is one half of a λ/2 dipole's field.	69
203	**A**	U0 depends on loop area, number of windings and the angle θ to the transmitter.	71
204	**B**	A more gradual widening and a longer funnel give more gain.	73
205	**B**	It makes the beams at the centre and the edge leave in phase.	76
206	**C**	With the correct delay, all beams leave the lens at the same time, they are in parallel.	76
207	**C**	It is formed by a slot in a sufficiently sized metal sheet or by a slot in a waveguide.	81
208	**A**	The dipole excites the parasitic elements which then also start transmitting.	67
209	**A**	Due to the distance to the dipole a phase shift is created.	67
210	**C**	On the sides the fields partly cancel, so gain is reduced.	67
211	**C**	The rod antenna is used wherever the mechanical dimensions must be as small as possible.	69
212	**B**	The loop acts like a coil, so the magnetic part of the field induces the voltage.	71
213	**B**	The propagation velocity of an electromagnetic wave depends on εr.	76
214	**C**	Dish: single direction, single beam. Lens: multiple direction, multiple beams.	78
215	**A**	Like a converging lens, they turn spreading energy into plane waves.	79

#	Ans	Why (LTT wording)	Page
216	**A**	The director is λ/2 − 5 %; the reflector is λ/2 + 5 %.	68
217	**B**	The field above the plane stays the same, even without the lower half.	69
218	**C**	The maximum voltage is induced when the loop plane matches the direction of propagation.	71
219	**A**	A flare under 40° with length and width large compared to λ.	73
220	**C**	Beams through the centre of the lens are delayed the most.	76
221	**B**	The lens material makes the electrical path longer, which delays the wave.	79
222	**C**	The field across the slot runs horizontally, so the wave is horizontally polarized.	81

9 · Antenna Characteristics and Antenna Tuning

#	Ans	Why (LTT wording)	Page
223	**B**	Lobe width is an indication of the directivity of an antenna.	83
224	**C**	The reference antenna is usually an isotropic antenna.	83
225	**A**	Normally, antenna gain is given in decibels.	83
226	**A**	More concentration means more field strength in that direction for the same power.	83
227	**B**	Gain is the extra radiated power in a direction compared to a reference antenna.	83
228	**C**	Antenna gain is a performance indicator gauged in comparison to a reference source.	85
229	**A**	Mechanical tuning adjusts the antenna length to the wavelength.	86
230	**B**	The antenna tip can be L or T shaped or be fitted with an "umbrella".	86
231	**C**	The length is matched to the frequency, either mechanically or electrically.	88
232	**B**	This diagram is the locus.	88
233	**B**	In order to tune the antenna to the frequency desired the reactance must be zero.	88
234	**A**	The HF coupler tunes the antenna over its frequency range.	89
235	**C**	The resonant frequency is reduced, i.e. the antenna is electrically "lengthened".	86
236	**B**	The angle where the radiated power has dropped to half its maximum.	83
237	**B**	This value is equivalent to the 3 dB point with respect to the main direction.	83
238	**A**	The use of a directional antenna has the same effect as an increase in transmitter power.	83
239	**B**	In practice, λ/2-dipole antennas are often used as reference antennas.	83
240	**A**	Boresight is the axis of maximum gain / maximum radiated power.	85
241	**A**	An isotropic antenna radiates equally everywhere, so its gain is 0 dBi.	85
242	**C**	On the X-axis the ohmic resistance is indicated, on the Y-axis the reactance.	88
243	**C**	A λ/2 dipole has a gain of 1.64 over an isotropic antenna.	83
244	**A**	Long wave wavelengths can be kilometres long, too long for a λ/4 antenna.	86
245	**A**	CE is in parallel with CA, so the total capacitance is larger.	86
246	**C**	f1: antenna too short, tuned by an inductor in series; the antenna is lengthened.	88

#	Ans	Why (LTT wording)	Page
247	**B**	f2: antenna too long, shortened by a capacitor in series.	88
248	**B**	f3: antenna too short, lengthened by a capacitor in parallel.	88
249	**C**	f4: antenna too long, shortened by an inductor in parallel.	88

10 · RF Lines, Matching and Standing Waves

#	Ans	Why (LTT wording)	Page
250	**A**	RF lines carry RF energy with as little loss as possible.	90
251	**C**	Tuned resonant circuits, resistance transformation and matching.	90
252	**C**	Coax is an asymmetric line: a shield around a central inner wire.	90
253	**B**	Typical impedances of symmetric twin cables start at approx. 150 Ω.	94
254	**A**	It is the power transfer theorem, or Jacobi's Law.	96
255	**A**	Electrically: symmetric and asymmetric. Physically: shielded and unshielded.	90
256	**B**	Both lines have equal capacity to ground and carry equal, opposite signals.	90
257	**B**	The asymmetric cable has one line on RF potential and the other line grounded.	90
258	**C**	The physical construction and the dielectric (insulation) material.	92
259	**B**	Line inductance L' and capacity C' determine the ratio of ix and ux.	92
260	**B**	In general, the impedance of a cable is calculated as Z0 = √(L/C).	94
261	**C**	They have higher dampening, lower voltage ratings and lower maximum power.	94
262	**C**	The impedances of transmitter and antenna must match.	96
263	**A**	A non-matching antenna creates a standing wave on the line.	98
264	**A**	SWR shows how far the antenna impedance differs from the line impedance.	98
265	**C**	A ratio of 1 means that the impedances are identical; we have maximum power transfer.	98
266	**B**	With open or shortened lines the ratio is infinite.	98
267	**A**	Reflections can be in-phase or opposite phase.	99
268	**C**	With a shorted or open ended line large standing waves for current and voltage appear.	102
269	**A**	One line capacity (C20) to ground is missing because ground itself is the conductor.	90
270	**C**	The inner wire has a capacity C12 against the shield, but no capacity C10 to ground.	90
271	**B**	A short or an open line gives a hard reflection; small mismatches give soft ones.	98
272	**B**	With an open line there is no current flow at the end but the voltage reaches its maximum.	100
273	**C**	With a shortened line the maximum current flows at the end but the voltage is zero.	100
274	**A**	A shorted λ/4 line (or open λ/2) acts as a parallel resonant circuit.	102
275	**A**	An open λ/4 line (or shorted λ/2) acts as a series resonant circuit.	102
276	**C**	Lower impedances are possible with co-axial cables where 50–75 Ω are common.	94

#	Ans	Why (LTT wording)	Page
277	**B**	The standard impedance of coaxial power cables in modern installations is 50 Ω.	94
278	**A**	Inductance L', capacity C', series resistance and parallel resistance.	92
279	**B**	Z0 is the voltage/current ratio of an infinitely long line.	94
280	**B**	A line is matched when RL = Z0.	97
281	**C**	The reflected wave runs back and both waves superpose; they may amplify or cancel out.	98
282	**A**	The greater the ratio the greater the deviation between both the impedances.	98
283	**B**	SWR = Emax / Emin = Imax / Imin.	98
284	**C**	SWR = Zload / Z0 or Z0 / Zload, whichever is greater.	98
285	**C**	High voltage can cause arcing and sparking that damages the dielectric.	100
286	**B**	High current heats spots half a wavelength apart, causing burns or small fires.	100
287	**A**	The in-phase peaks are half a wavelength apart.	100
288	**B**	A shorted λ/4 line or an open λ/2 line behaves as a parallel resonant circuit.	102
289	**C**	At the input the current has a node and the voltage has a loop.	102

11 · Resonant Circuits

#	Ans	Why (LTT wording)	Page
290	**A**	A mass pulled back by a return force, such as a spring or gravity.	105
291	**A**	Electric oscillator systems consist of inductors and capacitors.	105
292	**B**	Selective amplification, muting or transformation of frequencies.	116
293	**B**	The inductor stores it as a magnetic field, the capacitor as voltage.	105
294	**A**	UL and UC cancel, so a high current flows, limited only by RS.	109
295	**A**	The frequency range where the curve drops to a set value either side of the peak.	114
296	**C**	Bandwidth is determined by the quality factor Q.	114
297	**A**	The ability to suppress frequencies other than the resonant one.	114
298	**B**	The high current at f0 can make UL and UC much higher than the applied voltage.	109
299	**C**	iL and iC cancel, so only the current iR through RP flows.	111
300	**A**	A series circuit tuned to it shorts it to ground as the ground leg.	116
301	**C**	A parallel circuit tuned to it, as the ground leg, keeps only that frequency high.	116
302	**C**	L and C create a time delay between voltage and current.	105
303	**A**	The frequency is determined by the magnitude of the mass and the length l.	107
304	**B**	In a system without friction or other losses, their sum is constant at any instant.	107
305	**C**	An external voltage charges the capacitor.	107
306	**C**	The undamped oscillator frequency: f0 = 1 / (2π √(L C)).	107
307	**B**	Magnetic energy matches kinetic energy; electrical energy matches potential.	108
308	**A**	Their geometric sum is equal to the applied voltage U (2nd law of Kirchhoff).	109
309	**C**	Z is lowest at f0 because XL and XC cancel each other.	109

#	Ans	Why (LTT wording)	Page
310	**A**	As RS is a real load, the phase angle between U and i is zero degrees.	109
311	**B**	The current i is the geometric sum of the individual currents (1st law of Kirchhoff).	111
312	**B**	The apparent resistance Z equals RP at the resonant frequency.	111
313	**A**	A resistor is independent of frequency; its impedance is constant.	113
314	**C**	The impedance of an inductor is proportional to the frequency.	113
315	**A**	For a capacitor its impedance decreases (though not linearly) as the frequency increases.	113
316	**C**	At the intersection, XL = XC; that frequency is called the resonant frequency fR.	113
317	**B**	Normally, this is −3 dB (1/√2 = 0.707) from the peak value 1 (0 dB).	114
318	**A**	Bandwidth B = f0 / Q.	114
319	**C**	The magnetic field reaches its maximum strength when the voltage is 0.	107
320	**B**	The coil's magnetic field induces a voltage that recharges the capacitor reversed.	107
321	**C**	Below f0, UC is greater: capacitive. Above f0, UL is greater: inductive.	109
322	**A**	Below f0, iL is greater: inductive. Above f0, iC is greater: capacitive.	111
323	**B**	High selectivity usually costs bandwidth.	114
324	**B**	A large Q makes the curve sharp; a small Q makes it gradual.	115